

[Edit](#)

Generate Insights from Images by Using Azure AI Vision Services

Challenge Overview

Understand the scenario

You are a Developer for Hexelo, an organization that needs to generate insights from images by using Azure AI Vision Services. In this Challenge Lab, you will explore how to generate image captions by using Azure AI Vision Services. First, you will create a Computer Vision resource. Next, you will detect faces in an image, and then you will search photos by using image retrieval. Finally, you will add captions to images.

Navigating the Challenge Lab

🔗 ► Understanding a Cloud Slice

🔗 ► Quick tips for navigating the Challenge Lab instructions.

[New to Challenge Labs? Click here to learn more.](#)

Create a Computer Vision Resource

Hints Enabled

No

Yes

 Note About Evolving Technology

Azure AI technologies are constantly evolving as Microsoft continues to update, refine, and improve its tools and interfaces. As a result, you may notice slight differences between the instructions in this lab and what you see on your screen. Rest assured, we are actively working to keep all tutorials up to date with the latest features and interfaces. We appreciate your patience and understanding, and we apologize for any inconvenience this may cause.

- Sign into the virtual machine as **Admin** by using `!PasswOrd!` as the password.
- Open **Microsoft Edge**, go to `https://portal.azure.com`, sign in to the Azure portal as `{USERNAME}` by using `{PASSWORD}` as the password, and then dismiss all prompts.

 Expand this hint for guidance on signing into the Azure portal.

- On the Windows taskbar, in Search, enter **Edge**, and then select **Microsoft Edge** to open a new browser.
- In the Address bar, enter `https://portal.azure.com`, and then select **Enter**.
- In the Sign in dialog, enter `{USERNAME}`, and then select **Next**.
- In Enter password, enter `{PASSWORD}` and then select **Sign in**.
- Dismiss all prompts.

 Want to learn more? Review the documentation on [the Azure portal](#).

- Create a **Computer Vision** resource named `ComputerVision{LAB_INSTANCE_ID}` by using **Azure AI Services** and the values in the following table. For any property that is not specified, use the default value.

Property	Value
Resource group	<code>{RESOURCE_GROUP_NAME}</code>

Property	Value
Region	East US
Name	T ComputerVision{LAB_INSTANCE_ID}
pricing tier	Free F0
By checking this box I certify that I have reviewed and acknowledged all the terms above	Checked



You can ignore any warnings regarding verification of user roles.



Expand this hint for guidance on creating a Computer Vision resource.



Want to learn more? Review the documentation on [creating a Computer Vision resource](#).

- Obtain the **API Key** and **Endpoint** for the ComputerVision{LAB_INSTANCE_ID} resource, and then record them in the following **API Key** and **Endpoint** text boxes:

API Key

Endpoint



Expand this hint for guidance on locating your API Key and Endpoint.



Want to learn more? Review the documentation on [locating your API Key and Endpoint](#).

- Sign in to the Azure AI Vision Studio as

T

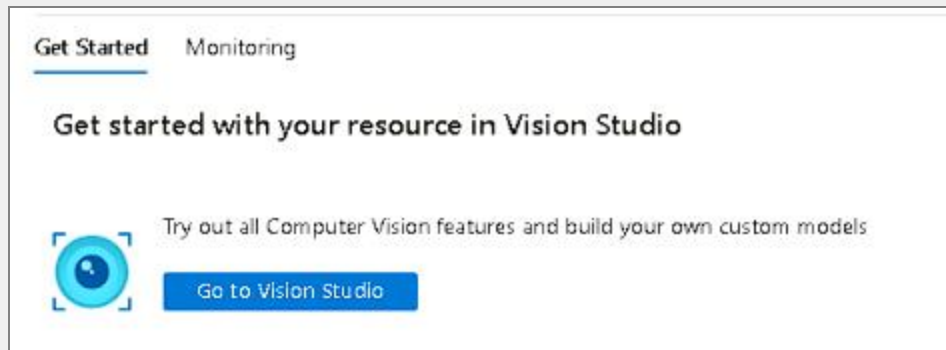
 {USERNAME} by using

T

 {PASSWORD} as the password

💡 Expand this hint for guidance on signing into the Azure AI Vision Studio. ^

- On the ComputerVision50191772 | Computer vision Overview page, on the Get Started tab, select **Go to Vision Studio**.



- On the Azure AI | Vision Studio Featured page, in the upper right-hand corner, select **Sign in**, and then sign in as **T {USERNAME}** by using **T {PASSWORD}** as the password.
- Close any welcome messages.

💡 Want to learn more? Review the documentation on [signing into the Azure AI Vision Studio](#).

Check your work

Verify

Detect faces in an image

Hints Enabled

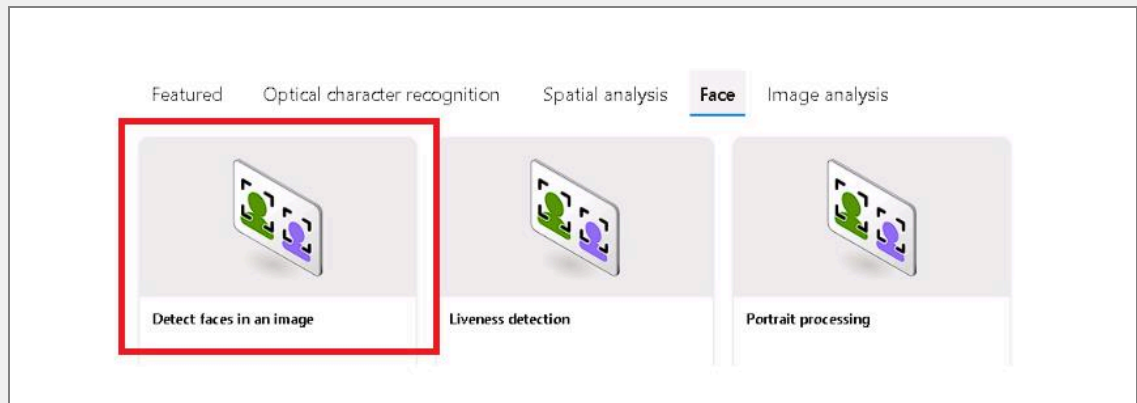
No Yes

- Create a Cognitive service resource by using the values in the following table. For any property that is not specified, use the default value.

Property	Value
Subscription	default
Name	T VisionResource{LAB_INSTANCE_ID}
Resource group	{RESOURCE_GROUP_NAME}
Region	East US
Pricing tier	S0
Azure Resource	VisionResource{LAB_INSTANCE_ID}

💡 Expand this hint for guidance on creating a resource.

- On the Azure AI | Vision Studio Featured page, in Get started with Vision, select the **Face** tab, and then on the Detect faces in an image card, select **Try it out**.



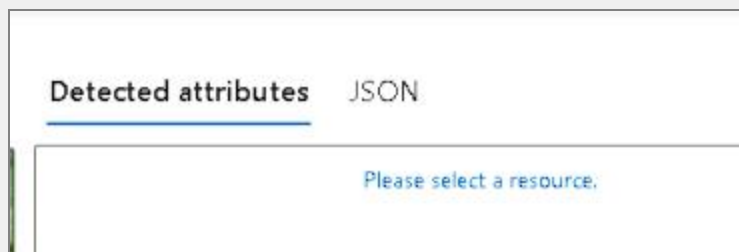
- On the **Detect faces in an image** page, in Try it out, select the checkbox to **verify the acknowledgement**.



- In *Use one of your own files or choose from a sample below*, select the first image.



- To the right of Sample image 1, on the Detected attributes tab, select **Please select a resource**.



- On the Select an Azure resource window, in Subscription, select the **default** subscription.

- In Azure Resources, select **Create a new resource**.



- On the Create a new resource window, in Name, enter `T`
`VisionResource{LAB_INSTANCE_ID}`, and then in Resource group, select `{RESOURCE_GROUP_NAME}`.
- In Location, verify the Location is set to **East US**, in Price tier, select **S0**, and then select **Create resource**.

A screenshot of a dialog box titled 'Select an Azure resource'. The dialog has a close button (X) in the top right corner. Inside, there is a paragraph of text: 'To access Vision Studio, you need a Cognitive Services resource. You can use an existing resource or create a new one. You can change the resource you use at any time. [Learn more about resources in Azure](#)'. Below this text, there is a section for 'Azure directory' with the URL 'cloudslice.onmicrosoft.com'. Then, there is a 'Subscription' section with a dropdown menu showing 'Skillable Original Content Dev 01'. Below the subscription, there is another 'Azure Resources' section with a search bar containing 'No resources found' and a blue link 'Create a new resource'. At the bottom right of the dialog are two buttons: 'Confirm' and 'Cancel'.

- On the Select an Azure resource window, verify that the Azure Resource is set to `VisionResource{LAB_INSTANCE_ID}`, and then select **Confirm**.

Select an Azure resource

To access Vision Studio, you need a Cognitive Services resource. You can use an existing resource or create a new one. You can change the resource you use at any time. [Learn more about resources in Azure](#)

Azure directory
cloudslice.onmicrosoft.com

Subscription *

Skillable Original Content Dev 01

Azure Resources * ⓘ

VisionResource50191772

[Create a new resource](#)

Confirm **Cancel**

💡 Want to learn more? Review the documentation on [signing into the Azure AI Vision Studio](#).

- Run Face Detection on Sample image 1, and then review the Detected attributes.

💡 Expand this hint for guidance on detecting faces in an image. ^

- On the **Detect faces in an image** page, in Try it out, reselect the checkbox to **verify the acknowledgement**.
- In *Use one of your own files or choose from a sample below*, select the first image.

Use one of your own files or choose from a sample below.



Drag and drop a file here
or
Browse for a file
or
Take a photo









ⓘ The information collected from your photo for this demo does not predict or classify facial attributes or create a facial template, nor is it used to identify you.

- To the right of Sample image 1, on the Detected attributes tab, review the **Detected attributes**.

💡 Want to learn more? Review the documentation on [signing into the Azure AI Vision Studio](#).

- Enter the Detected attribute and value, below the **Face #1** line for Sample Image 1 in the following text box:

Sample Image 1 Detected attributes

- Run Face Detection on Sample image 2, and then review the JSON output.

💡 Expand this hint for guidance on detecting faces in an image. ^

- On the **Detect faces in an image** page, in Try it out, if necessary, reselect the checkbox to **verify the acknowledgement**.
- In *Use one of your own files or choose from a sample below*, select the second image.



- To the right of Sample image 2, on the Detected attributes tab, review the **Detected attributes**.
- Select the **JSON** tab, review the output, and then enter the **faceRectangle: width** value in the textbox below.

💡 Want to learn more? Review the documentation on [signing into the Azure AI Vision Studio](#).

- Enter the width of Sample Image 1 in the following text box:

Sample Image 2 width

- Repeat the previous task by using **Sample image 3**, locate the JSON **noseAndMouthCovered** value for either face, and then enter it in the following text box:

JSON NoseAndMouthCovered value

- Repeat the previous task by using **Sample image 4**, and then review the Detected attributes and JSON output.

Check your work

A blue rectangular button with the word "Verify" in white text.

Search photos by using image retrieval

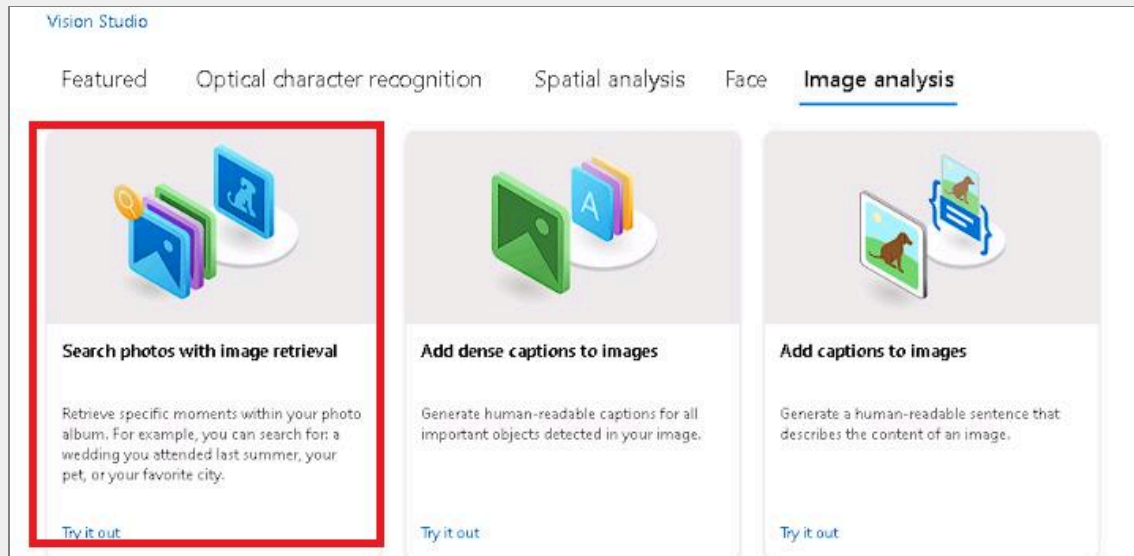
Hints Enabled

No Yes

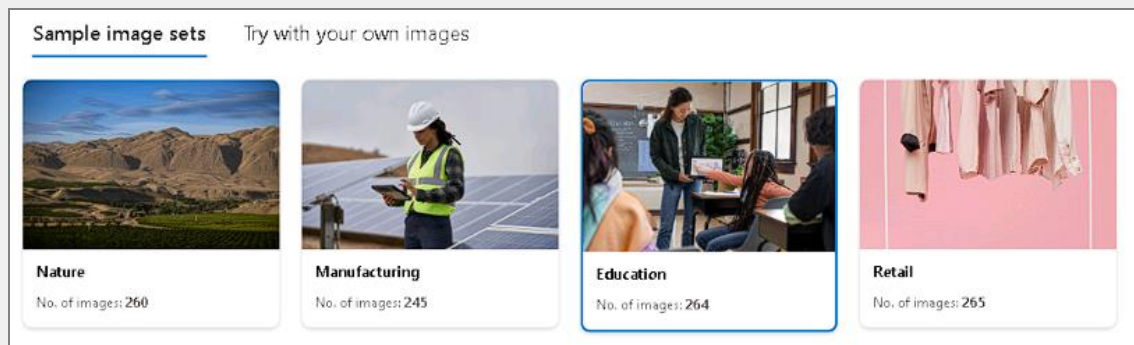
- Using the **Education** Sample image set, perform a search query for **Students using laptops**, and then review the **Most relevant** query results**

💡 Expand this hint for guidance on performing a retrieval query.

- Select **Vision Studio** to return to the Azure AI | Vision Studio Get started with Vision page.
- On the Get started with Vision page, select the **Image analysis** tab, and then, on the *Search photos with image retrieval* card, select **Try it out**.



- On the Search photos with image retrieval page, on the Sample image sets tab, select the **Education** sample image set.



- In *Select a retrieval query or create your own*, select **Students using laptops**, and then select **Search**.



- In Query results, drag the slider all the way to the left, towards **Most relevant**, and then record the total number of images in the following textbox.

Query results

Query results vary from most relevant in the dataset to least relevant. Utilize the slider below to view more or less images based on their relevance to the retrieval query.

Most relevant Least relevant

1



2



3



4



 Want to learn more? Review the documentation on [the Azure portal](#).

- Record the number of **Most relevant** images from the **Education** sample image set, in the following text box:

Education image set: Most Relevant images

- Using the **Retail** Sample image set, perform a search query for **Black leather jackets**, and then review the **Most relevant** query results.
- Record the number of **Most relevant** images from the **Retail** sample image set, in the following text box:

Retail image set: Most Relevant images

Check your work

Verify

Add captions to images

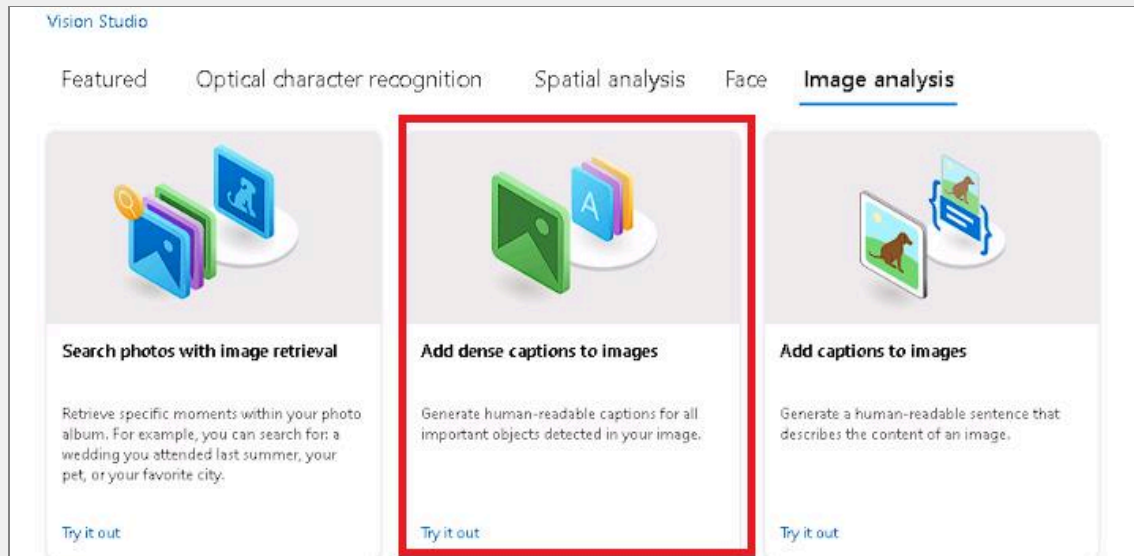
Hints Enabled

No Yes

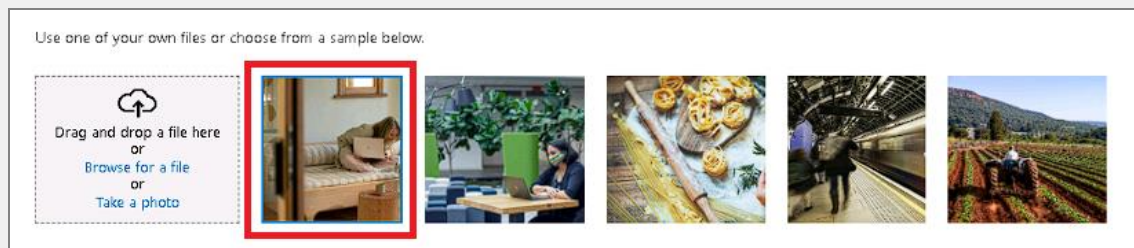
- Add dense captions to **Sample image 1**, and then review the Detected attributes.

💡 Expand this hint for guidance on adding dense captions to images. ^

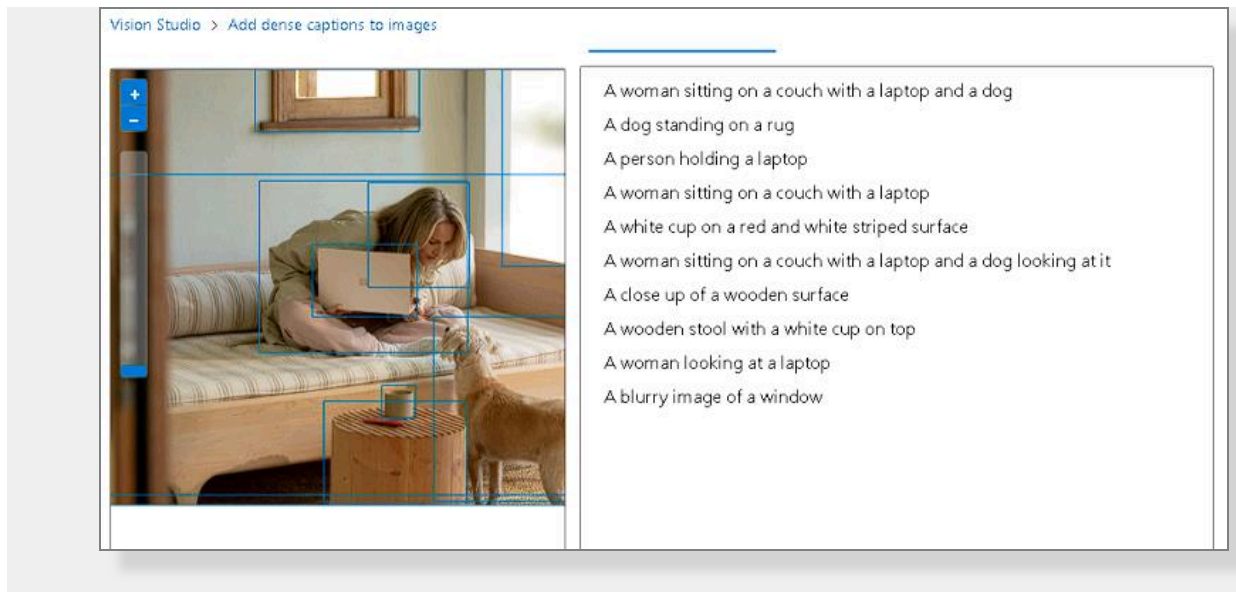
- Select **Vision Studio** to return to the Azure AI | Vision Studio Get started with Vision page.
- On the Get started with Vision page, select the **Image analysis** tab, and then, on the *Add dense captions to images* card, select **Try it out**.



- In *Use one of your own files or choose from a sample below*, select the first sample image.



- On the Detected attributes tab, review the dense captions.



 Want to learn more? Review the documentation on [signing into the Azure AI Vision Studio](#).

- Enter the number of generated dense captions for sample image 1 in the following textbox:

Sample image 1: Dense Caption count

- Add dense captions to **Sample image 1**, and then review the Detected attributes.

💡 Expand this hint for guidance on adding dense captions to images.

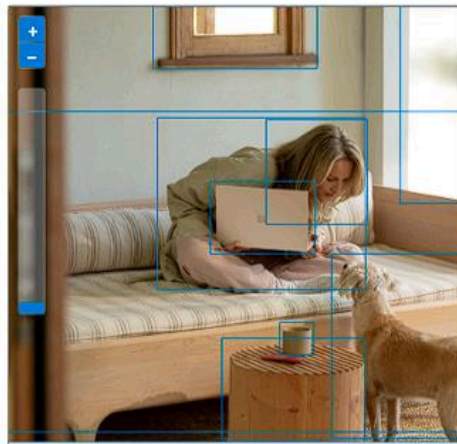
- In *Use one of your own files or choose from a sample below*, select the last sample image.

Use one of your own files or choose from a sample below.



- On the Detected attributes tab, review the dense captions.

Vision Studio > Add dense captions to images



A woman sitting on a couch with a laptop and a dog
A dog standing on a rug
A person holding a laptop
A woman sitting on a couch with a laptop
A white cup on a red and white striped surface
A woman sitting on a couch with a laptop and a dog looking at it
A close up of a wooden surface
A wooden stool with a white cup on top
A woman looking at a laptop
A blurry image of a window

💡 Want to learn more? Review the documentation on [signing into the Azure AI Vision Studio](#).

- Enter the last dense caption for sample image 5 in the following textbox:

Sample image 5



Detected attributes JSON

A man driving a tractor in a farm
A man driving a tractor
A blurry image of a tree
A man riding a tractor
A blue sky above a mountain
A tractor in a field

Sample image 5: Last Dense Caption

- Add dense captions to **Sample image 3**, and then locate the **confidence** score for the **a person holding a sprig of rosemary** caption.

💡 Expand this hint for guidance on adding dense captions to images. ^

- In *Use one of your own files or choose from a sample below*, select the third (middle) sample image.
- On the Detected attributes tab, review the dense captions.
- Select the JSON tab, and then on line 10, locate the **confidence** score.

- Enter the confidence score for the **a person holding a sprig of rosemary** caption in the following textbox:

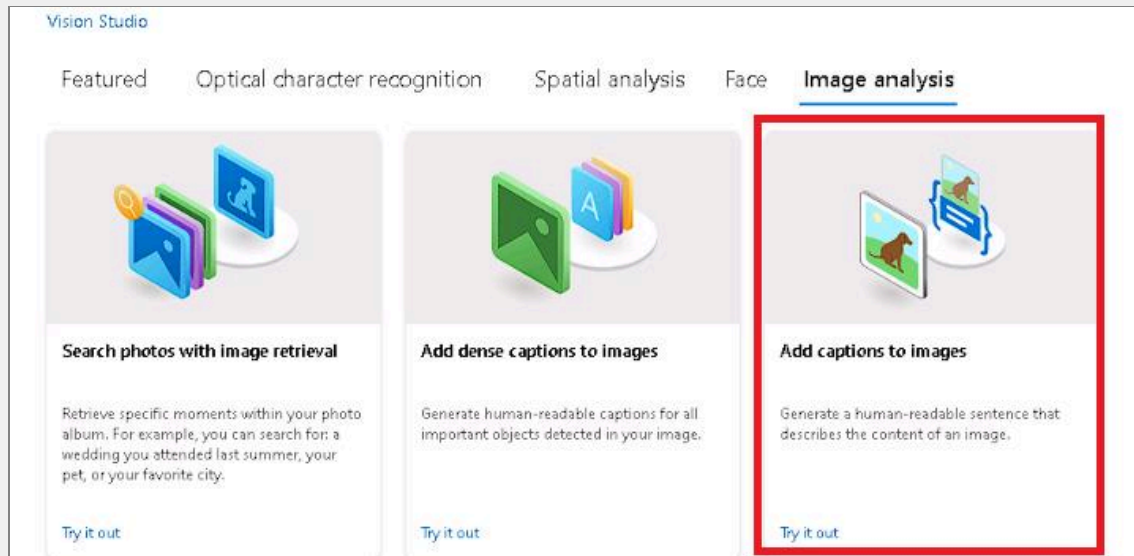
```
denseCaptionsResult: {  
  "values": [  
    {  
      "text": "a person holding a sprig of rosemary",  
      "confidence": 0.7674311995506287,  
      "boundingBox": {  
        "x": 0,  
        "y": 0,  
        "x2": 100,  
        "y2": 100  
      }  
    }  
  ]  
}
```

Confidence Score

- Add a one sentence caption to **Sample image 2**, and then review the Detected attributes.

💡 Expand this hint for guidance on adding captions to images.

- Select **Vision Studio** to return to the Azure AI | Vision Studio Get started with Vision page.
- On the Get started with Vision page, select the **Image analysis** tab, and then, on the *Add captions to images* card, select **Try it out**.



- In *Use one of your own files or choose from a sample below*, select the second sample image.



- On the Detected attributes tab, review the Detected attributes.

💡 Want to learn more? Review the documentation on [signing into the Azure AI Vision Studio](#).

- Enter the caption for sample image 2 in the following textbox:

Sample image 2: Caption

- Enter the caption confidence score for sample image 2 in the following textbox:

Sample image 2: Confidence Score

Check your work

Verify

Summary

Congratulations, you have completed the **Generate Insights from Images by Using Azure AI Vision Services** Challenge Lab.

You have accomplished the following:

- Created a Computer Vision resource.
- Detected faces in an image.
- Searched photos by using image retrieval.
- Added captions to images.

Ending your lab

To ensure your lab is recorded as complete, select **Submit** or **End** below. Exiting [X] the lab will result in an incomplete status.

 Once you select **Submit** or **End**, you will not be able to return to this Challenge Lab.

Your feedback is important!

As you end your Challenge Lab, please take a few minutes to complete the short survey that will appear in the next window. Alternatively, you may provide your feedback directly to [Challenge Labs feedback](#).

Looking for your next Challenge Lab?

Below are recommended Challenge Labs to try next.

- Explore Document Processing in Azure AI's Document Intelligence Studio [Guided]
- Create a Bot by Using the Bot Framework SDK [Guided]
- Generate Insights from Images by Using Azure AI Vision Services [Guided]
- Build an AI Model in Azure AI Machine Learning Studio [Guided]
- Integrate Azure AI Immersive Reader Into Your Application by Using Its SDK [Guided]
- Summarize Text Automatically by Using Azure AI Language Services [Guided]

- Translate Text Between Languages by Using Azure AI Translator [Guided]
- Generate Image Descriptions from Text by Using Azure AI Services [Guided]
- Build an AI-Powered Search System by Using Azure AI Search [Guided]
- Analyze and Generate Text-Based Sentiment by Using Azure AI Services [Guided]

